

Chance B. Loveday

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Education:

Maryville College, Maryville TN	May 2026
B.S. in Mathematics	GPA: 3.92
B.A. in Computer Science	
Minors: Analytics, Data Analysis, and Statistics	

Research Appointments:

Senior Thesis Research: CAD EMG-Controlled Robotic Prosthetic Hand	Sep 2025 – Apr 2026
➤ Pursued a capstone project focused on the design, manufacturing, and control of a functional, low-cost prosthetic hand. ➤ Designed several prototypes with dynamic range of motion in CAD. ➤ Fabricated the mechanical structure using high-performance additive manufacturing techniques and advanced materials (ASA). ➤ Developed and integrated a tendon-actuation system utilizing servo motors for biologically accurate grasping kinematics.	
OMNI Intern Oak Ridge National Laboratory	Jun - Aug 2024
➤ Built, trained, and fine-tuned an AI model on a visual transformer architecture to identify malignant cells such as tumors from MRI scans. ➤ Utilized the PyTorch and Monai libraries to write Python scripts that handle JSON databases. ➤ Wrote bash scripts to implement HPC resources available via the Frontier supercomputer.	
SULI Intern Oak Ridge National Laboratory	May - Jul 2023
➤ Developed an intuitive tool to create objects from JSON databases and generate radiation transport code inputs for various modeling systems using C++. ➤ Developed scripts that adhere to methods of quality assurance.	

Professional Certificates & Training:

3D Printing and Additive Manufacturing Certificate University of Illinois	Oct 2025
➤ Completed a 3D printing specialization covering manufacturing techniques, industry applications, software (Tinkercad, Fusion 360), materials, printer operations, culminating to a capstone project of designing and producing a functional 3D-printed object	
High-Performance Computing Crash-Course Certificate OLCF	Sep 2023
➤ Covered the basics of Vim, Python, C, multiprocessing programming, and machine learning on the Ascent cluster	

Technical Skills:

Programming & Analysis: Python (PyTorch), C++, C#, JavaScript, SQL, HTML/CSS
CAD and Design: Fusion 360, Tinkercad
Advanced Manufacturing: FDM/SLA 3D printing, laser cutting and engraving, slicing (Ultimaker Cura, Bambu Studio), machine maintenance and troubleshooting
Shells & CLI: Bash, Vim
Research Tools: LaTeX, R, High-Performance Computing (HPC)

Professional Presentations:

Loveday, C.*, Jorge Estrada Martinez, Jesse G. Smith 2026. *Exploring Design and Additive Manufacturing of Low-Cost Prosthetic Hands*. Poster Presentation, 17 April 2026, Maryville College, Maryville, TN 37804

Loveday, C.*, Isaac R. Lyngaas 2024. *Building a 3D Neuroimaging Deep-Learning Model With a Vision Transformer*. Poster Presentation, 31 July 2024, Oak Ridge National Laboratory, Oak Ridge, TN 37830

Loveday, C.*, Douglas E. Peplow, and Robert A. Lefebvre 2023. *Design and Implementation of a Radiation Transport Input Code Generator Interface*. Poster Presentation, 26 July 2023, Oak Ridge National Laboratory, Oak Ridge, TN 37830

Additional Experience:

<i>STEM Education Summer Research Assistant and Mentor Maryville College</i>	<i>Aug 2023 - Aug 2025</i>
➤ Provided mentorship to cohorts of first-year STEM students during a two-week summer bridge program. ➤ Facilitated a Micro Bit coding lab to promote technical problem-solving and coding logic in 2023 and 2024. ➤ Created and facilitated an activity teaching students the fundamentals of CAD modeling via Tinkercad and slicing software in collaboration with the MC3D printing club in 2025.	
<i>TRIO STEM Tutor Maryville College</i>	<i>Sep 2024 - Apr 2025</i>
➤ Provided support in computer science, mathematics, and statistics for TRIO students coming from first-generation and low socio-economic backgrounds.	
<i>STEM Success Lead Academic Mentor Maryville College</i>	<i>Sep 2023 - May 2026</i>
➤ Managed tutoring center operations and offered support in computer science, mathematics, and statistics.	
<i>Teaching Assistant Maryville College</i>	<i>Aug 2023 - May 2026</i>
➤ Provided in-class support, led lectures, and gave supplemental instruction for seven distinct courses. ➤ Courses led include Calculus I, Differential Equations, Probability & Statistics I, Probability & Statistics II, Introduction to Computer Programming, Data Structures, and Algorithms.	
<i>Adventure Group Facilitator Mountain Challenge LLC</i>	<i>Mar 2023 - Oct 2024</i>
➤ Manage Maryville College and outside groups of adults and children through outdoor team-building exercises while ensuring the safety of all participants.	

Awards & Recognition:

<i>George A. Knapp Mathematics Award Maryville College</i>	<i>Apr 2026</i>
<i>CCSC Regional Programming Contest 2nd Place Mercer University</i>	<i>Nov 2025</i>
<i>Russell N. & Frances Lane Edwards & Troy Bell Lane Waggoner Math Award</i>	<i>Apr 2025</i>
<i>CCSC Regional Programming Contest 4th Place Furman University</i>	<i>Nov 2024</i>
<i>West Chevrolet Computer Science Award Maryville College</i>	<i>Apr 2024</i>
<i>Hall Ethical Thinking Essay Contest 2nd Place</i>	<i>Apr 2024</i>
<i>Regional Programming Contest Award 2nd Place Lipscomb University</i>	<i>Apr 2023, 2025</i>
<i>Dean's List Maryville College</i>	<i>Dec 2022 - May 2026</i>
<i>Computer Science Club President & Co-Founder</i>	<i>Sep 2022 - May 2026</i>
<i>MC3D Printing Club Treasurer, President</i>	<i>Sep 2022 - May 2026</i>

Community Outreach Projects:

<i>Union Grove Elementary 3D Printing Demo</i>	<i>Oct 2025</i>
<i>GP High School Bio Lab Demo</i>	<i>Mar 2025</i>
<i>Sevier County High School STEM Recruitment</i>	<i>Mar 2025</i>
<i>CSC Website Team Lead</i>	<i>Sep 2024 - Present</i>
➤ Launched a custom website to promote Maryville College's Computer Science Club and provide computer-science-related resources to the public community (https://csclub.maryvillecollege.edu/). ➤ Built from scratch using HTML, CSS, JavaScript, and Flask.	